Supporting Information for "Advancing Seasonal Water Supply Forecasting for Lakes and Reservoirs Using Copulas"

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Additional Supporting Information (Files uploaded separately)

1. Caption for Dataset S1

Introduction

The supporting information includes results for analyses discussed in the main manuscript for different spatial and time scales than are shown in the main manuscript. Additionally, the data and code needed to reproduce our analyses are described here and included as additional uploaded files for the review process, but will be uploaded to a publicly available repository prior to potential acceptance.

Data Set S1. Dataset one contains all data and code necessary to reproduce the results shown in both the main manuscript and supporting information. The dataset consists of input data in .csv format, scripts in .R format, intermediate output in .RDS format, and figures in .png format. Base input data, found in the 'L2SWBM_med_1950_2022' folder, is reformatted by the 'InputDataFormatting.R' and stored in the 'formatted_input' folder. The copula models are then fit to this data by the 'FittingandSampling.R' script, which fits the distributions and performs random sampling. The 'MatchingAndPlotting.R' script performs 'matched' sampling and all figure generation. Intermediate data objects, including model distributions and samples, once generated, are stored in the 'R_objects' folder.

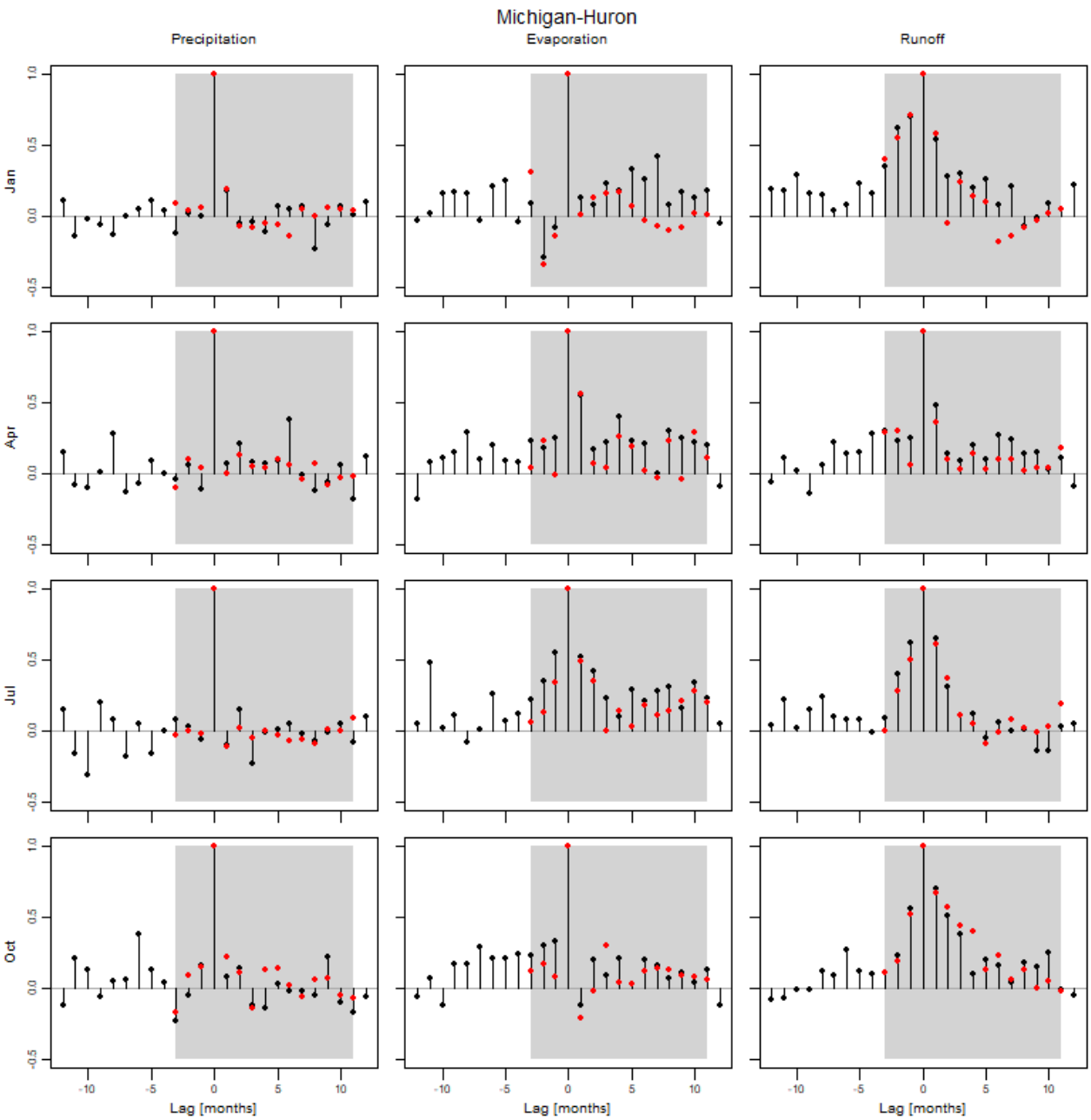

Figure S1. Serial correlation in monthly Lake Michigan-Huron water balance components from historical data (black) and copula simulations (red). For clarity, only select months, representative of each season, are shown. The grey region highlights a 15-month calibration window which includes a 3-month 'pre-forecast' period for matching initial basin conditions followed by a 12-month forecast horizon.

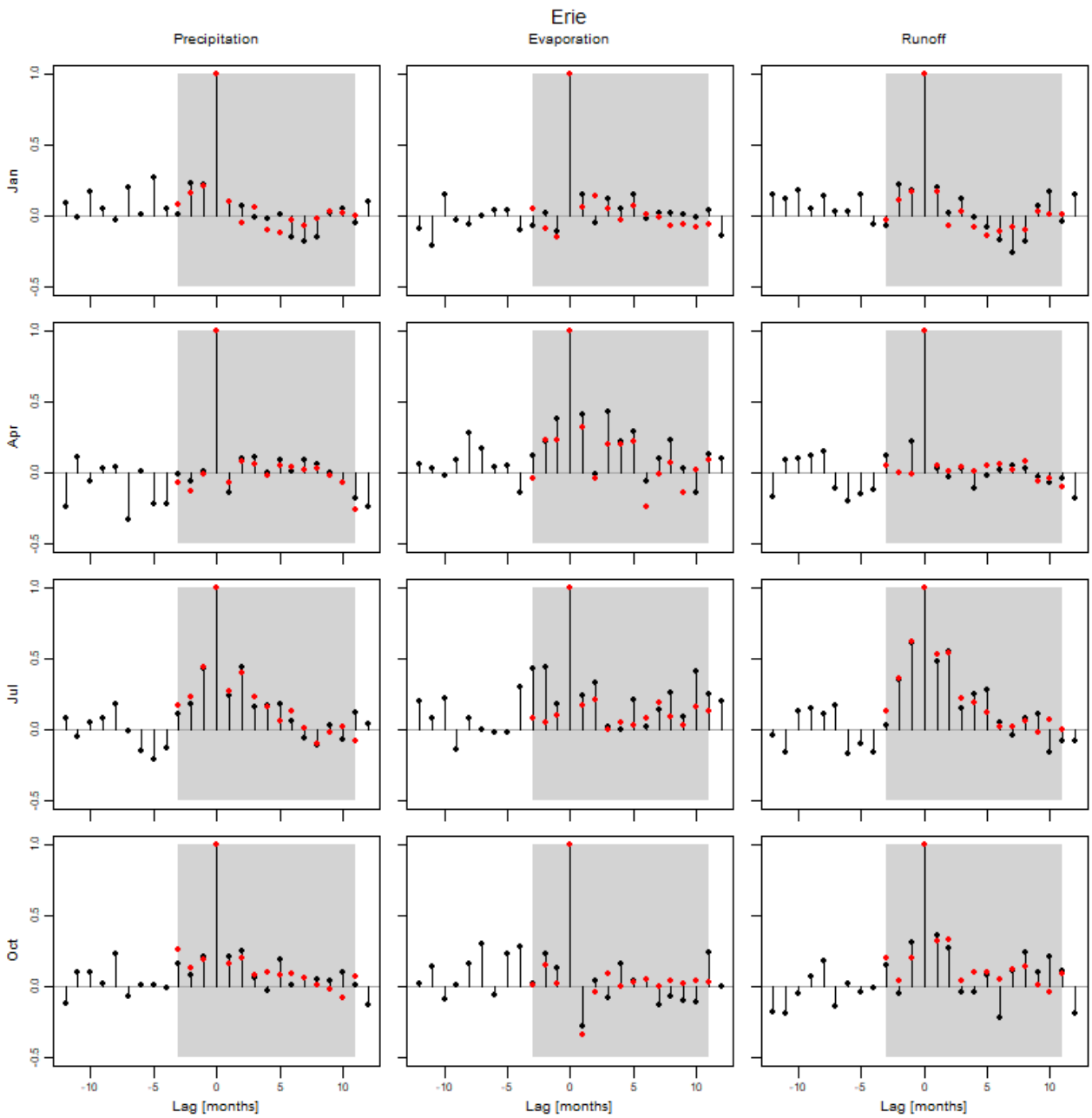

Figure S2. Serial correlation in monthly Lake Erie water balance components from historical data (black) and copula simulations (red). For clarity, only select months, representative of each season, are shown. The grey region highlights a 15-month calibration window which includes a 3-month 'pre-forecast' period for matching initial basin conditions followed by a 12-month forecast horizon.

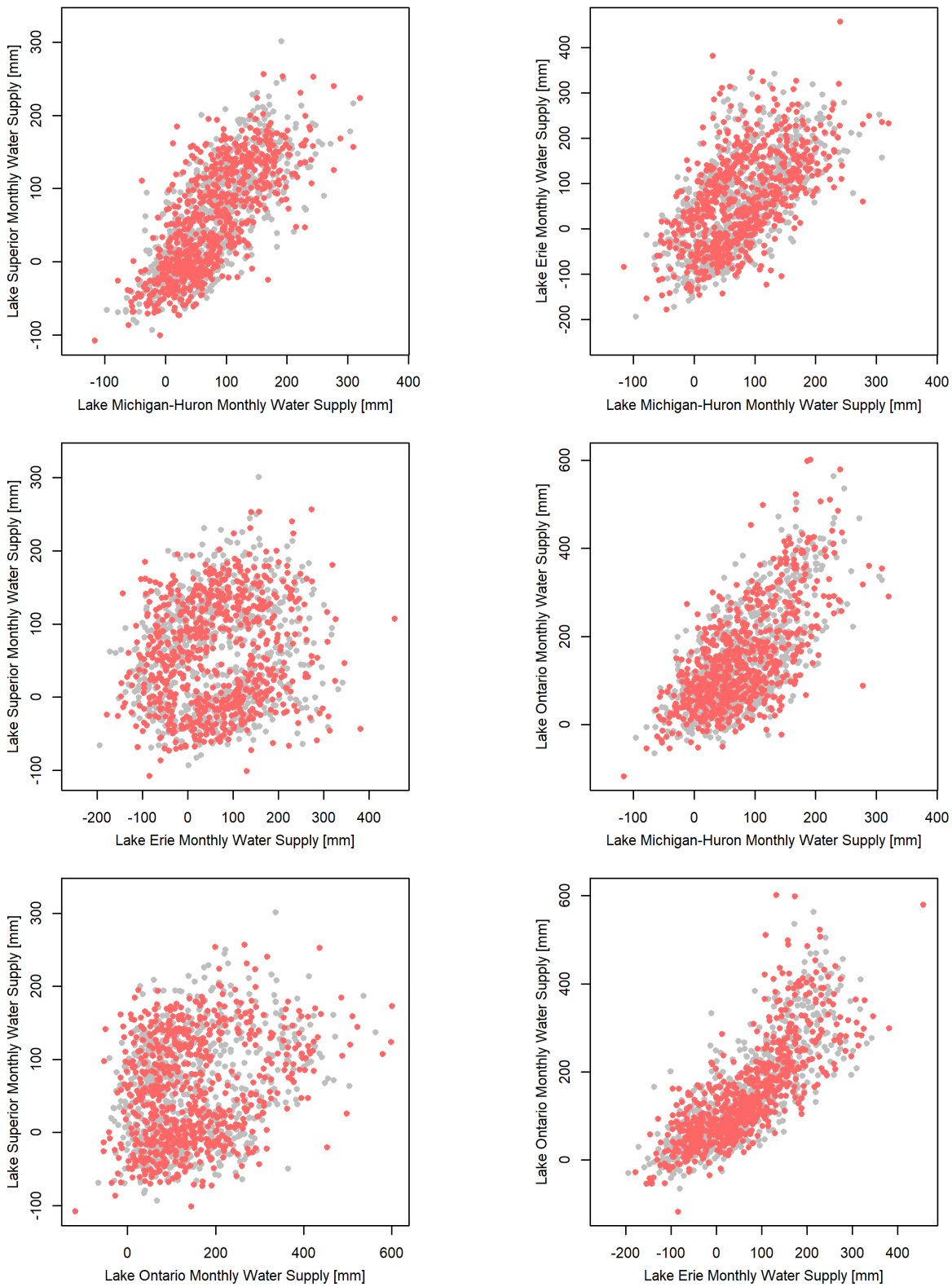

Figure S3. Spatial correlation in monthly Great Lakes water supplies from historical data (grey) and copula simulations (red).

June 8, 2024, 4:36pm

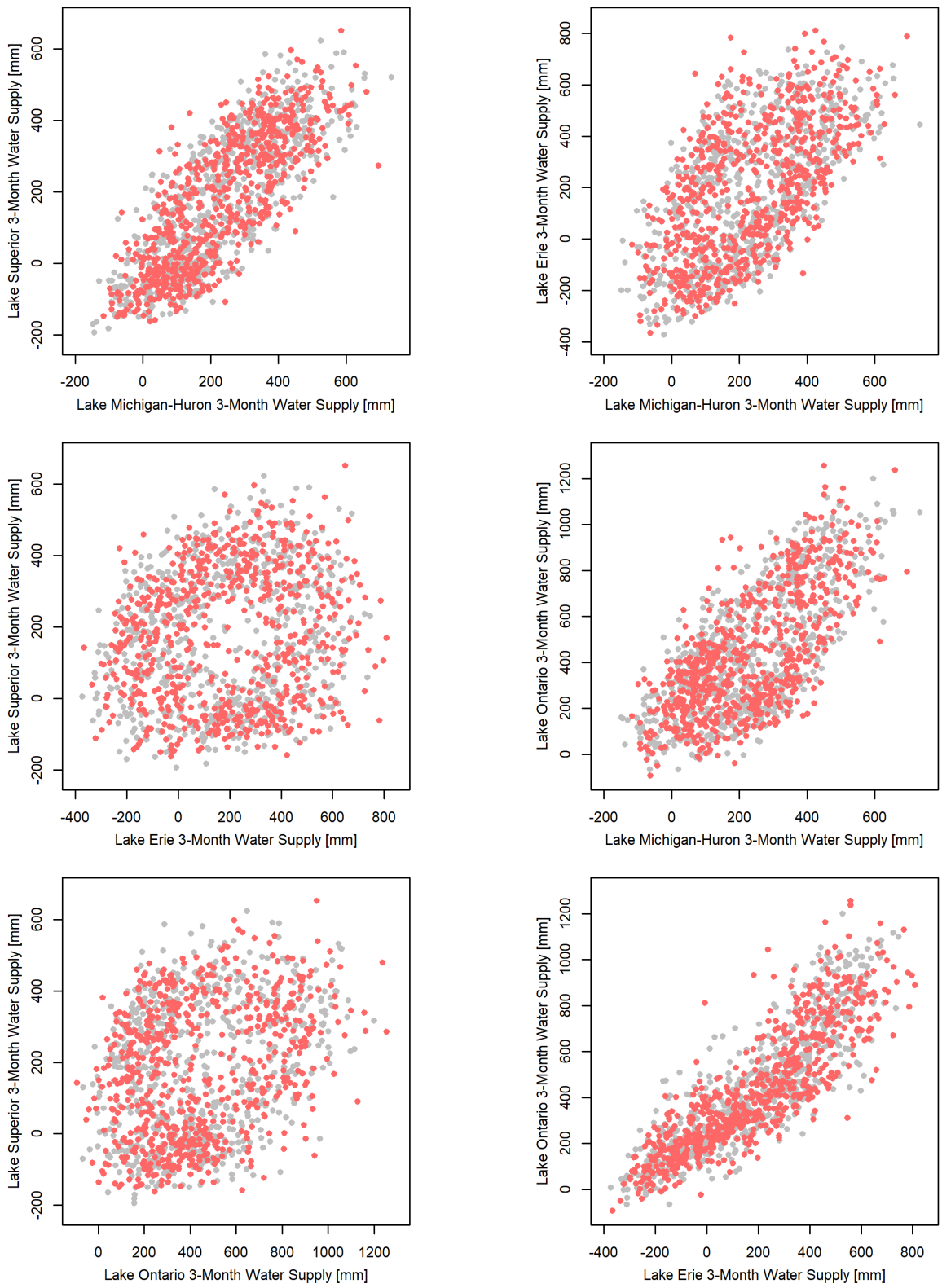

Figure S4. Spatial correlation in 3-month Great Lakes water supplies from historical data (grey) and copula simulations (red).

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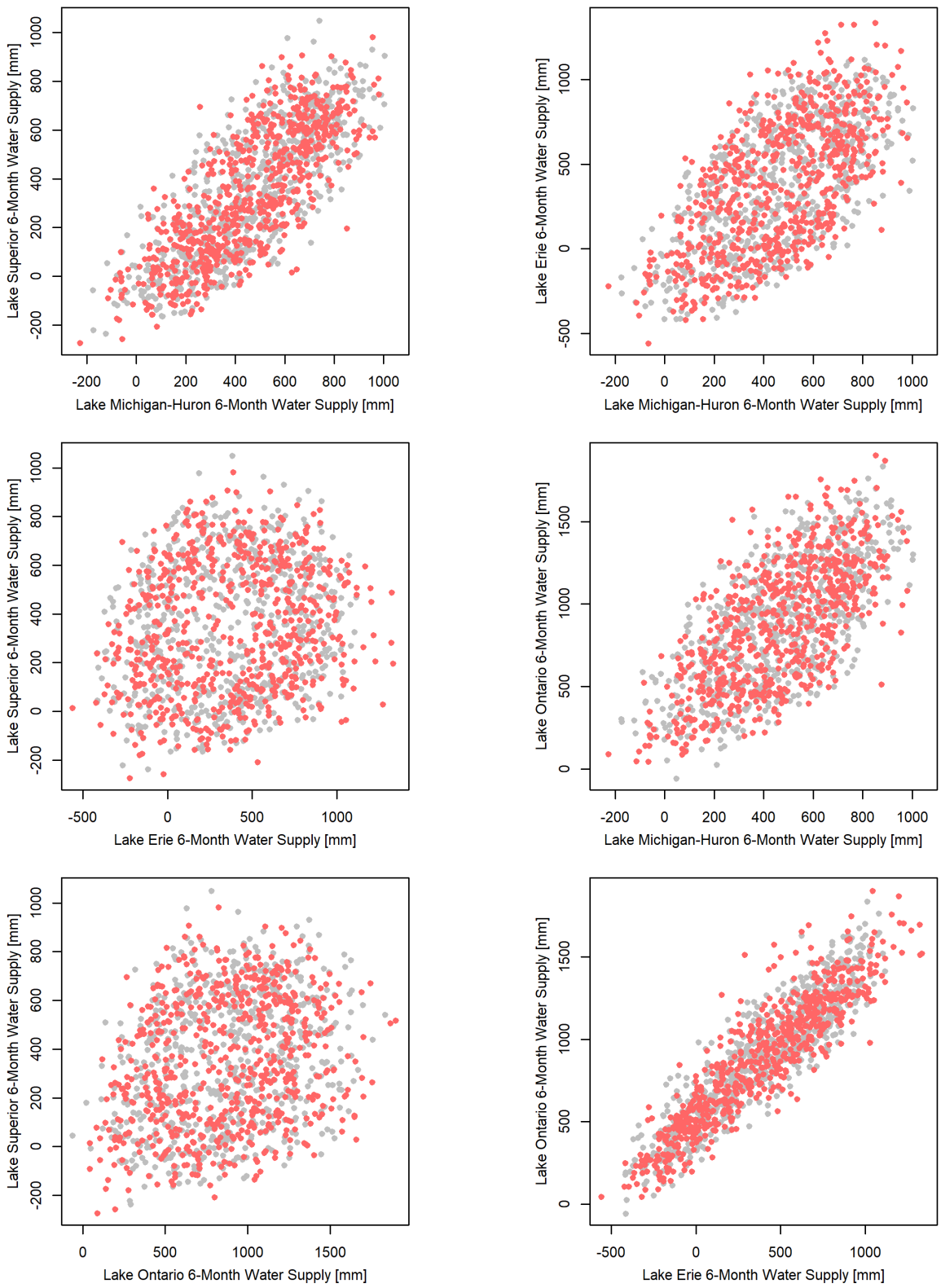

Figure S5. Spatial correlation in 6-month Great Lakes water supplies from historical data (grey) and copula simulations (red).

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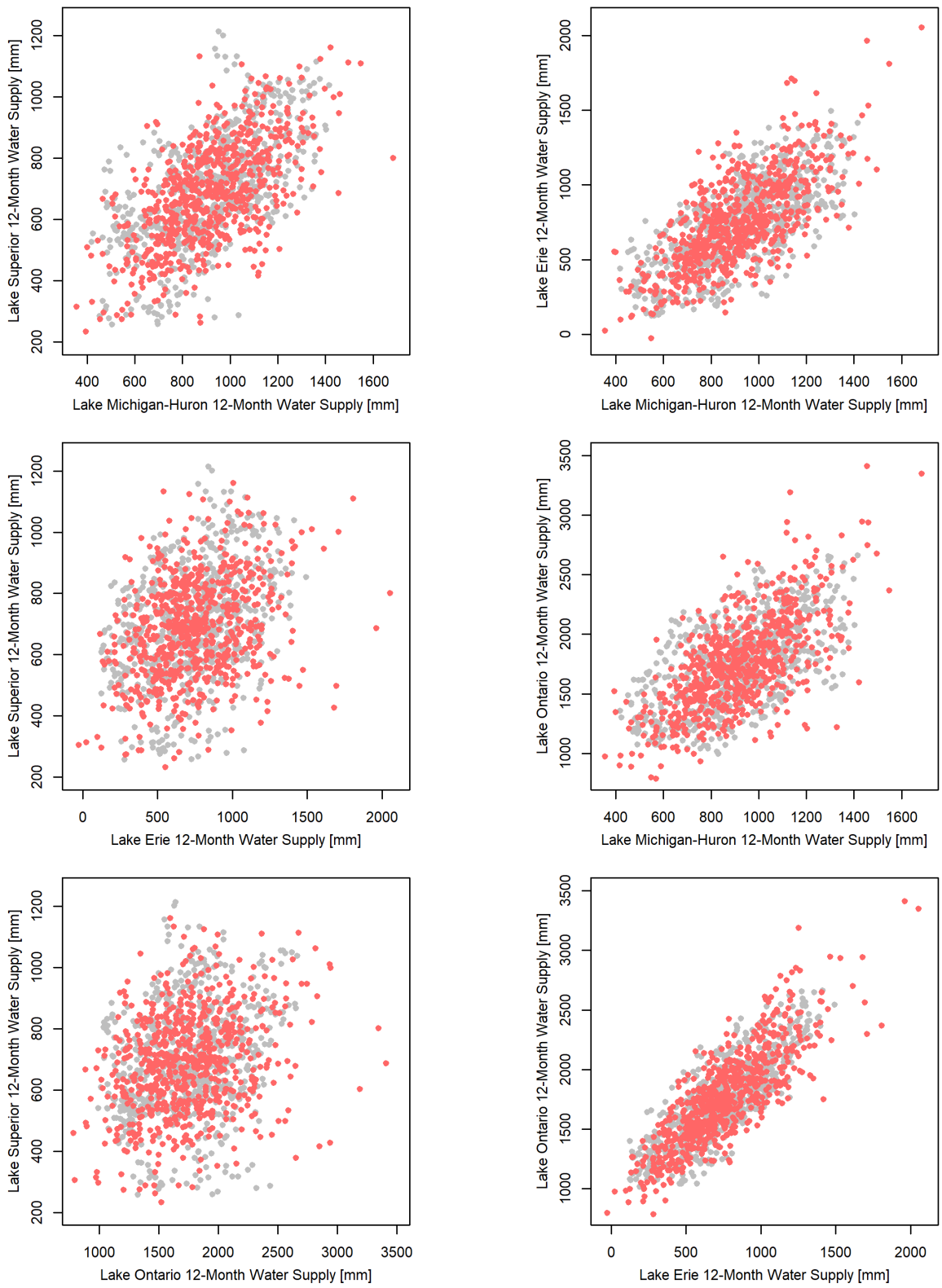

Figure S6. Spatial correlation in annual Great Lakes water supplies from historical data (grey) and copula simulations (red).

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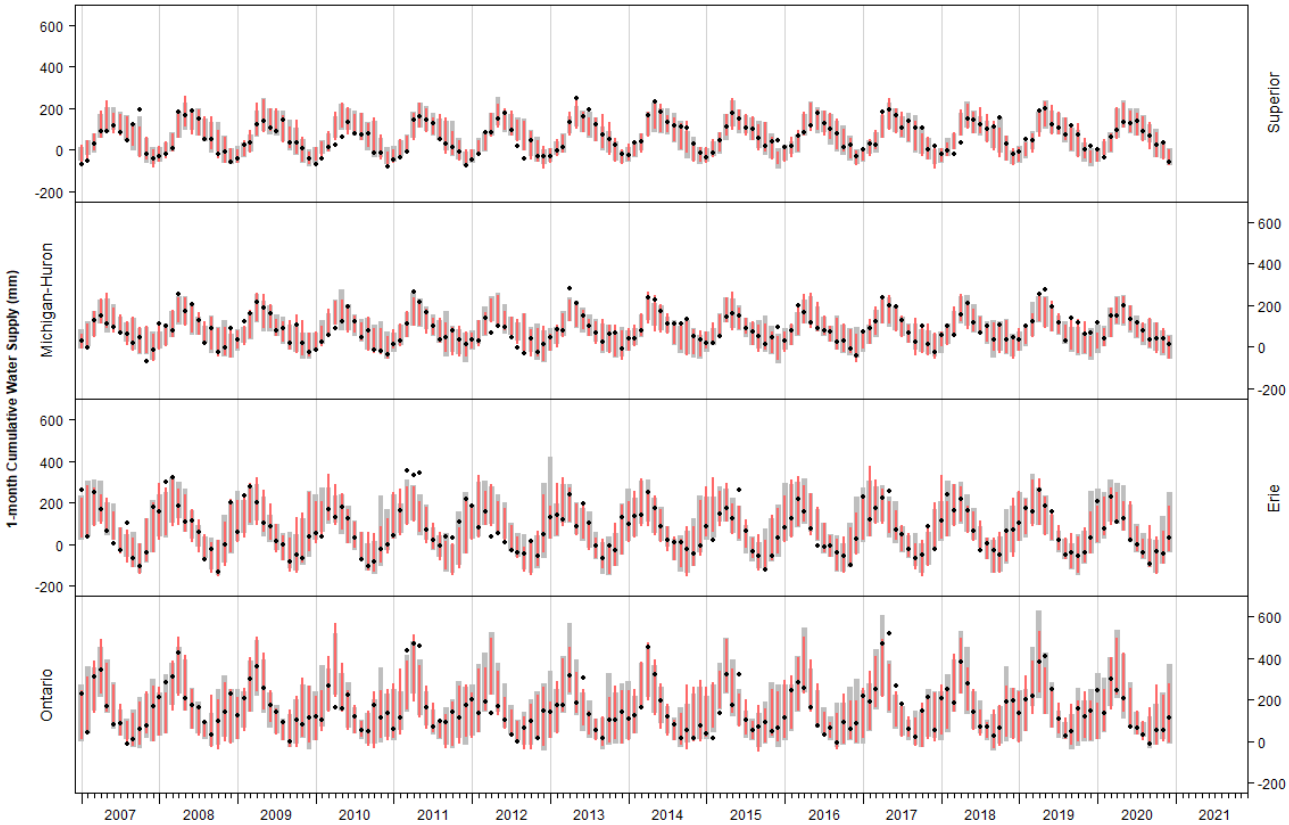

Figure S7. Comparison between observed (black dots) and simulated (grey and red vertical bands) 1-month cumulative water supplies across all of the Great Lakes from Jan 2007 through Dec 2020. Simulated cumulative water supplies are presented as 90% credible intervals for ensembles encoded with (red) and without (grey) matching to initial conditions.

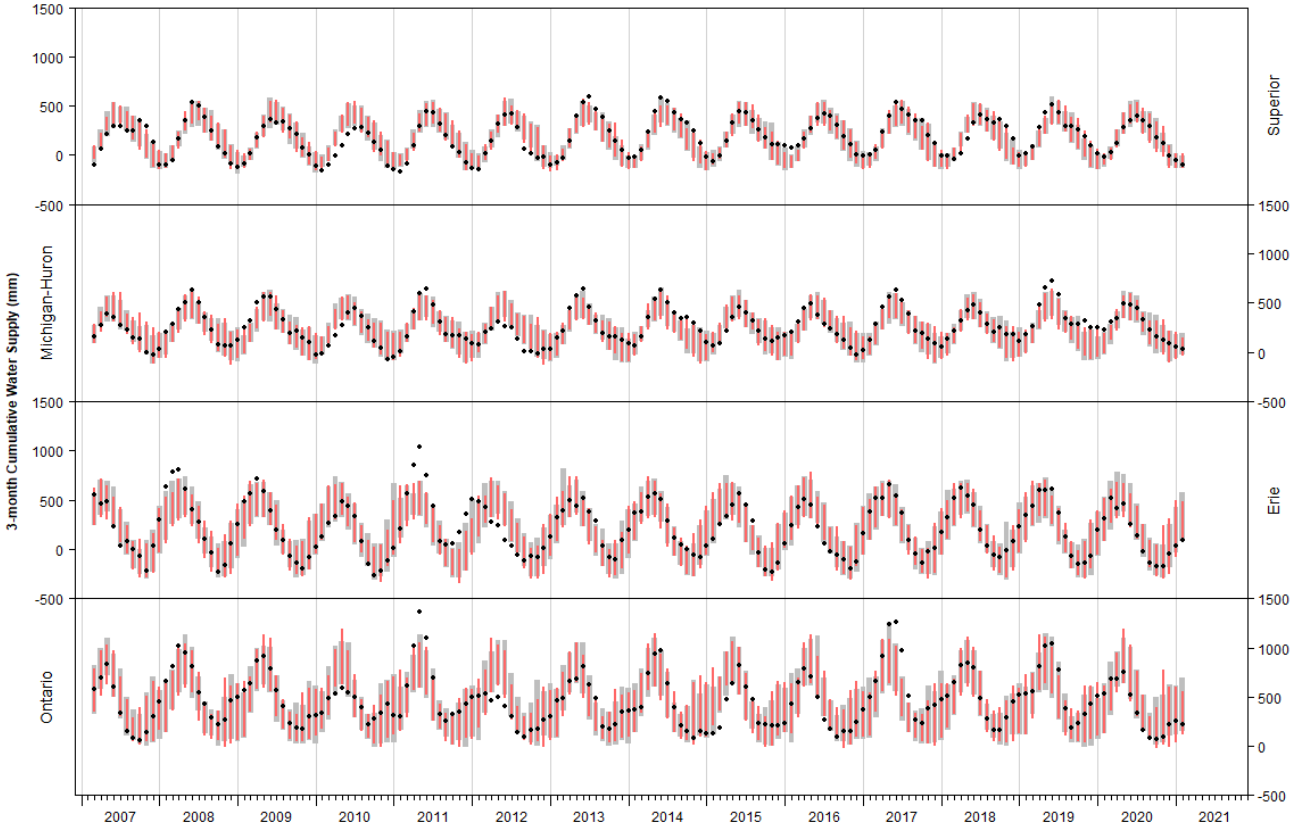

Figure S8. Comparison between observed (black dots) and simulated (grey and red vertical bands) 3-month cumulative water supplies across all of the Great Lakes from March 2007 through February 2021 (months represent the end of each 6-month cumulative supply period). Simulated cumulative water supplies are presented as 90% credible intervals for ensembles encoded with (red) and without (grey) matching to initial conditions.